Practical : 7

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Problem Statement: Divide and Conquer (Merge Sort) - CO1, CO2, CO5

Sort Online Orders by Delivery Time using Merge Sort. Problem Statement: You are given a list of online orders, each with an estimated delivery time in minutes. Write a program to sort these orders using the Merge Sort algorithm so the delivery system can prioritize quicker deliveries first.

Code

```
# Merge Sort


def mergeSort(arr):


    if len(arr) > 1:


        mid = len(arr) // 2

        left = arr[:mid]

        right = arr[mid:]


        mergeSort(left)

        mergeSort(right)


        i = j = k = 0


        # Merge the two halves

        while i < len(left) and j < len(right):
```

```
        if left[i] < right[j]:
            arr[k] = left[i]
            i  += 1
        else: arr[k] =
            right[j]
            j  += 1


            k += 1

    # Copy remaining elements
    while i < len(left):
        arr[k] = left[i]
        i          +
        = 1k
        += 1

    while j < len(right):
        arr[k] = right[j]
        j          +
        = 1k
        += 1



# Main Program


orders = []
```

```
n = int(input("Enter number of online orders: "))

for i in range(n): time = int(input("Enter delivery
    time (minutes): ")) orders.append(time)

print("\nOriginal Delivery Times:")
print(orders)

mergeSort(orders)

print("\nSorted Delivery Times (Quickest First):")
print(orders)
```

Output

Input

Enter number of online orders: 6

Enter delivery time (minutes): 45

Enter delivery time (minutes): 20

Enter delivery time (minutes): 60

Enter delivery time (minutes): 15

Enter delivery time (minutes): 35

Enter delivery time (minutes): 25

Output

Original Delivery Times:

[45, 20, 60, 15, 35, 25]

Sorted Delivery Times (Quickest First):

[15, 20, 25, 35, 45, 60]